

Exponent Laws with Variable BasesQuestion Bank - Additional 3.T1 **ANSWER KEY**

*Show your work. Circle your final answer.***For questions 1 through 6, write an equivalent expression as a single monomial. Assume no variable equals 0.**

1. $y^6 \cdot y^7$

y^{13}

2. $(k^3)^8$

k^{24}

3. $(5x^2)^3$

$125x^6$

4. $\frac{b^{11}}{b^4}$

b^7

5. $(-3a^5)(-4a^2)$

$12a^7$

6. $\frac{18x^6y^4}{6x^5y}$

$3xy^3$

For questions 7 and 8, apply more than one law. Write each answer as a single monomial.

7. Write an equivalent expression.

a. $(3a^4b^2)(6ab^3)$

$18a^5b^5$

b. $(2p^3q^5)^3$

$8p^9q^{15}$

c. $(xy^3)^2 \cdot x^4y$

x^6y^7

8. Write an equivalent expression.

a. $\frac{(2x^3y)^4}{8x^7y}$

$2x^5y^3$

$16x^{12}y^4$ over $8x^7y$.

b. $\frac{m^5n^6}{m^2n^6}$

m^3

$n^0 = 1$.

c. $\frac{(5c^2)^2}{5c^3}$

$5c$

$25c^4$ over $5c^3$.

Working backwards.

9. Find the value of n that makes each statement true.

a. $a^n \cdot a^5 = a^{12}$

$n = 7$

b. $(x^3)^n = x^{18}$

$n = 6$

c. $\frac{y^n}{y^4} = y^6$

$n = 10$

$n - 4 = 6$.

10. A rectangle has length $9a^5$ and width $3a^2$.

a. Write its area as a single monomial.

$27a^7$

b. A square has side length $7b^3$. Write its area as a single monomial.

$49b^6$

11. Which expression is equivalent to $(4y^2)^3$?

☐ A. $12y^6$

☐ B. $4y^6$

☒ C. $64y^6$

☐ D. $64y^5$

C

12. Select ALL of the expressions that are equivalent to m^{10} .

☒ **A.** $m^3 \cdot m^7$

☒ **B.** $(m^2)^5$

☐ **C.** $m^2 \cdot m^5$

☐ **D.** $\frac{m^{20}}{m^2}$

☒ **E.** $\frac{m^{14}}{m^4}$

☒ **F.** $(m^5)^2$

A, B, E and F